

Building a Medical Expert System Using Knowledge Representation.

Introduction

An expert system is a computer program designed to mimic the decision-making ability of a human expert in a specific domain. It captures specialist knowledge in a structured form and uses reasoning to draw conclusions from that knowledge.

This lab develops a simple medical expert system for a hospital setting. The system takes a patient's reported symptoms as input and applies a set of IF–THEN rules to identify possible illnesses.

Objectives:

- Represent medical knowledge using facts and rules
- Design and implement a rule-based inference engine
- Apply forward chaining to derive diagnoses from symptoms
- Visualize the knowledge structure using a semantic network diagram

Knowledge Base Design

The knowledge base is stored in a JSON file and contains four components: facts, symptoms, diseases, and rules.

Symptoms

The system recognises ten symptoms:

No.	Symptom	No.	Symptom
1	Fever	6	Runny Nose
2	Headache	7	Fatigue
3	Cough	8	Sore Throat
4	Chest Pain	9	Vomiting
5	Sneezing	10	Diarrhea

Diseases

The system can diagnose four conditions: Malaria, Pneumonia, Flu, and Food Poisoning.

Rules

The four IF–THEN production rules are:

Rule	IF Conditions	THEN Diagnosis
R1	Fever AND Headache AND Fatigue	Malaria
R2	Cough AND Chest Pain AND Fatigue	Pneumonia
R3	Sneezing AND Runny Nose AND Sore Throat	Flu
R4	Vomiting AND Diarrhea AND Fatigue	Food Poisoning

Note that Fatigue appears in rules R1, R2, and R4. This means a patient reporting Fatigue alongside other symptoms may trigger more than one rule, resulting in multiple candidate diagnoses.

Rule-Based Inference System

System Design

The inference engine is implemented in Python and follows a three-layer design:

- **Knowledge Layer** — the JSON file stores all symptoms, diseases, and rules
- **Inference Layer** — a function loads the knowledge base and applies forward chaining
- **Interface Layer** — a command-line menu accepts symptom input and displays results

How the Inference Works

1. The patient's reported symptoms are loaded into a working memory set
2. The engine iterates through every rule in the knowledge base
3. For each rule, it checks whether ALL conditions in the IF part appear in working memory
4. If all conditions match, the rule fires and the disease is recorded as a possible diagnosis
5. All fired rules are returned and displayed to the user.

Worked Example

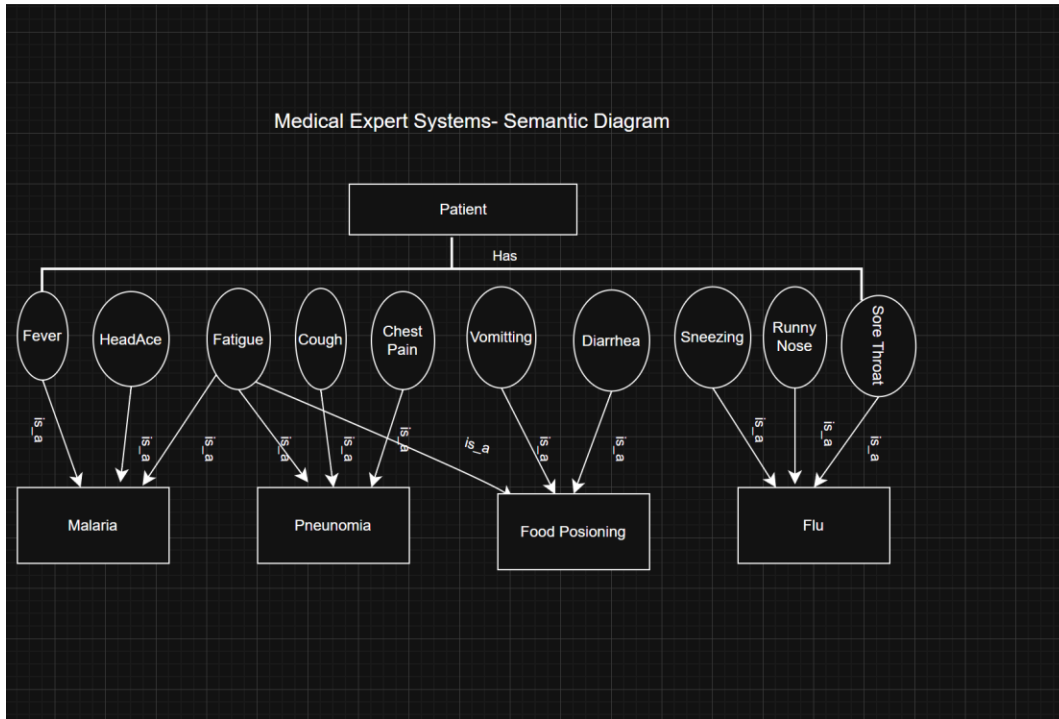
```
=== Medical Expert System ===
Enter patient name: Levi

Enter symptoms separated by commas
Symptoms: Runny Nose, Sneezing, Sore Throat

Patient: Levi
Possible illness: Flu
PS C:\Users\Public\KR_ Practical_Lab1> 
```

Semantic Network Diagram

A semantic network is a graphical representation of knowledge where nodes represent concepts and labelled directed edges represent relationships between them.



Conclusion

This lab successfully demonstrated how medical knowledge can be encoded explicitly using facts and IF-THEN rules, and how a forward-chaining inference engine can apply that knowledge to derive diagnoses from patient-reported symptoms. The system correctly identifies all four target diseases when the appropriate symptom combinations are provided.

The semantic network diagram makes the relationships between patients, symptoms, and diseases visually clear, reinforcing the conceptual model behind the implementation.

